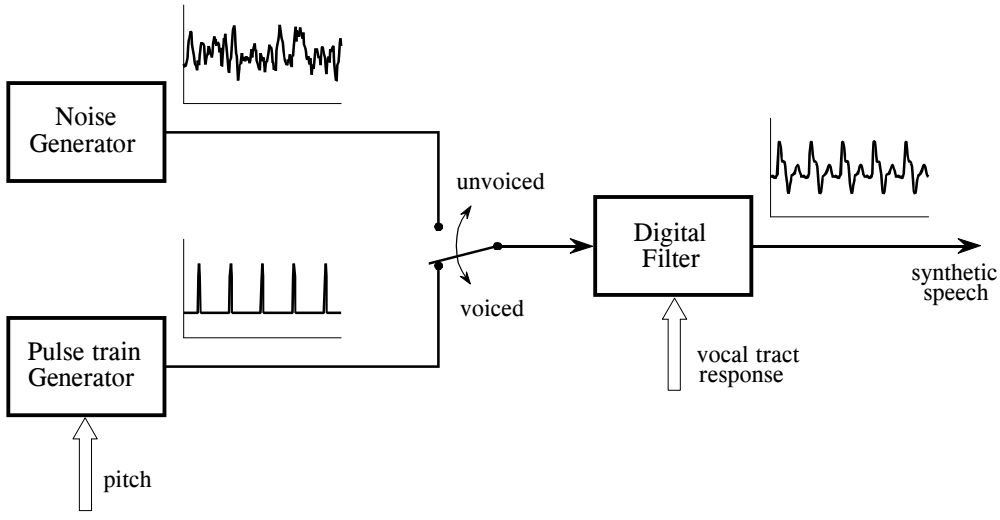

FIGURE 22-8
Human speech model. Over a short segment of time, about 2 to 40 milliseconds, speech can be modeled by three parameters: (1) the selection of either a periodic or a noise excitation, (2) the pitch of the periodic excitation, and (3) the coefficients of a recursive linear filter mimicking the vocal tract response.

the air flow, just behind the Adam's apple. In response to varying muscle tension, the vocal cords vibrate at frequencies between 50 and 1000 Hz, resulting in periodic puffs of air being injected into the throat. Vowels are an example of voiced sounds. In Fig. 22-8, voiced sounds are represented by the pulse train generator, with the pitch (i.e., the fundamental frequency of the waveform) being an adjustable parameter.

In comparison, *fricative* sounds originate as random noise, not from vibration of the vocal cords. This occurs when the air flow is nearly blocked by the tongue, lips, and/or teeth, resulting in air turbulence near the constriction. Fricative sounds include: *s*, *f*, *sh*, *z*, *v*, and *th*. In the model of Fig. 22-8, fricatives are represented by a *noise generator*.

Both these sound sources are modified by the acoustic cavities formed from the tongue, lips, mouth, throat, and nasal passages. Since sound propagation through these structures is a linear process, it can be represented as a linear filter with an appropriately chosen impulse response. In most cases, a *recursive* filter is used in the model, with the recursion coefficients specifying the filter's characteristics. Because the acoustic cavities have dimensions of several centimeters, the frequency response is primarily a series of resonances in the kilohertz range. In the jargon of audio processing, these resonance peaks are called the **format frequencies**. By changing the relative position of the tongue and lips, the format frequencies can be changed in both frequency and amplitude.

Figure 22-9 shows a common way to display speech signals, the **voice spectrogram**, or **voiceprint**. The audio signal is broken into short segments,